

Chapter 9: The "Brains" (Intro to JavaScript)

I. Static vs. Dynamic & Client vs. Server

- The Job of JS: JavaScript is the "brains" or "_____ " that makes a website _____ (interact, react, and change).
- Analogy: HTML = _____ (Nouns). CSS = _____ (Adjectives). JS = _____ (Verbs).

Static vs. Dynamic

Site Type	Description	Key Feature
Static	Like a poster; content is _____ and does not change.	Every user sees the exact same page.
Dynamic	Like a robot; _____ to user actions and changes its content (e.g., likes, search results).	The content is generated based on user input or a database.

The Two Brains of the Web

Scripting Type	Where the Code Runs	What it is Good For
Client-Side (JS)	Runs directly inside the user's _____.	Instant _____, form validation, image rollovers.
Server-Side (PHP, Python)	Runs on the web _____ (a faraway computer).	_____ logins, saving data, accessing databases.

Deep Dive Discussion

1. **Form Validation:** Why is checking if a user's email *looks* valid a **client-side** job, but checking if the password is *correct* a **server-side** job?
2. **JS Placement:** When linking an external JavaScript file (<script src="...">), why is the recommended placement **right before the closing </body> tag**?

Application Activity

1. **File Structure:** Sketch or list the professional file structure for the "PixelPaws" project, including the main directory and the three internal folders.

II. Variables, Functions, and Logic

Variables: "Boxes" for Storing Data

- **let:** Used for variables that _____ (e.g., a score).
- **const:** Used for variables that _____ change (e.g., an HTML element or a player name).
- **Comments (//):** _____ that the browser completely ignores.

Functions & Parameters

- **Function:** A named, _____ block of code designed to perform a specific task (a "recipe").
- **Parameter:** The " _____ " you pass **into** a function when you define it.
- **return:** The keyword used to send a value _____ a function (the "finished dish").

Conditional Logic and Events

- **Conditional Logic:** The ability for code to " _____ " using if/else statements.
- **Event:** A signal that the user did something (e.g., _____, mouseover, submit).
- **addEventListener():** The command that tells JavaScript to _____ for an event on an element and run a function.
- **event.preventDefault():** A critical command that tells the browser to _____ (e.g., stopping a form from reloading the page).

Deep Dive Discussion

1. **Function Power:** Why is writing a function that uses a **parameter** (like function makeGreeting(petName)) more powerful than writing a function that doesn't use parameters?
2. **const vs. let:** Give an example of a website feature where you would *have* to use the let keyword and one where you would use const.

Application Activity

1. **Conditional Logic:** Write a simple if/else block that checks if a student's grade variable is passing (70 or higher). If it is, log "Pass"; otherwise, log "Fail."

- **Code:**

```
let grade = 75; // Imagine the user has 75%
```

```
// Your conditional logic starts here:
```

//

//

[illegible]

//

III. The Ultimate Magic Trick (Creating HTML from Data)

- **The DOM (Document Object Model):** The browser's internal _____ (or "family tree") of the HTML page.
- **querySelector():** The command used to _____ an element using a CSS selector.
- **classList.toggle():** A "magic" command that adds a CSS class if it's absent, and _____ it if it's present.

Organizing Data

Data Type	Symbol	Analogy	Description
Array	_____	The "Digital Backpack"	A variable that holds an _____ of items.
Object	_____	The "Trading Card"	Holds a collection of _____ _ pairs.

Loop	for...of	The "Auto-Repeater"	A command to repeat code for _____ item in an Array.
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Dynamic HTML Creation

These commands are used inside a loop to build HTML from scratch:

Command	What it Does
<code>document.createElement("div")</code>	Creates a new HTML element in _____.
<code>element.textContent</code>	Sets the _____ that appears inside the element.
<code>container.append(element)</code>	_____ the new element onto the page, inside the container.

Deep Dive Discussion

1. **Database Design:** Why is it more efficient to store the "PixelPaws" pet list as an **Array of Objects** instead of trying to store all the pet names in one Array, all the pet facts in another Array, and all the images in a third Array?
2. **Debugging:** What two tools/commands are essential for debugging JavaScript code, and why is the **Console** more helpful than just reloading the page?

Application Activity

1. **Object Access:** Given the pet object below, write the JavaScript code to log **only** the pet's fun fact to the console.

```
const pet = {
  name: "Mittens",
  type: "Cat",
  funFact: "Mittens once caught a laser dot."
};
```

- **Code:**
